

Chapter – 7: Particulate Nature of Matter

- Might collect pebbles, stones – riverbank, beach
- Mountains – rocks break – erosion – rivers flow – these regions – carry eroded rocks
- River flows – continue break down rocks – pebbles, stones, sand – transport large qty. – plains
- Bigger rocks – break down – finer grains – sand, clay – is this smallest unit OR breaks further?

What is Matter Composed of?

- Activity –
 - Take – chalk stick – break – 2 pieces
 - Continue breaking – until difficult – break further
 - Grind small pieces – mortar, pestle
 - **Observe** – fine chalk powder – magnifying glass
 - Each tiny grain – still chalk
- Recall – grade 7 – curiosity – changes around us – learnt – grinding chalk – physical change – no new substance
- Chalk powder – broken further – smaller particles – further grinding
- Reach stage – chalk particles – cannot break further – tiny units – basic building blocks
- Whole chalk – made of – large no. – smaller units – constituent particles
- **Constituent particles** – basic unit – makes up – larger piece – substance, material
- Grains – sand, clay – NOT – constituent particles – big rock – instead – made of constituent particles themselves
- **Explore** further – recall – dissolution of sugar into water
- Activity –
 - Fill – glass tumbler – drinking water
 - Add – 2 teaspoons sugar
 - Do not stir – taste spoonful water – top layer
 - Stir continuously – sugar dissolves
 - Again – taste spoonful water – top layer
- Notice – diff. – sweetness
- Top layer – sweet taste – after dissolving – present in solution
- Sugar particles – NOT obs. – BUT – presence – sensed – taste
- Sugar dissolves – breaks up – constituent particles – cannot break further
- Each tiny grain – sugar – millions constituent particles
- Above act. – support idea – matter – compose – large no. – extremely small particles
- Particles – so small – cannot be obs. – ordinary microscope
- Tiny sugar particles – separate – occupy – available space – water particles
- These spaces – between particles – **interparticle space**

What Decides Different States of Matter?

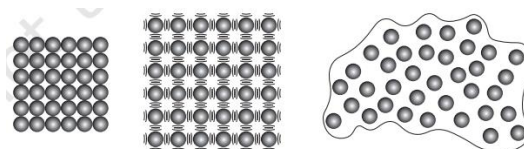
- Constituent particles – matter – held together – attractive forces – **interparticle attractions**
- Strength – attractions – depend – nature – substance – interparticle distance
- Slight increase – distance – drastic decrease – interparticle forces
- Strength – forces – decide – physical state – substances

- **Interesting fact –**

- Acharya Kanad – ancient Indian philosopher – idea – *Parmanu* (atom)
- Belief – matter – made of – tiny, indivisible particles – *Parmanu*
- This idea – noted – his work – *Vaisheshika Sutras*

Solid state

- Activity –
 - **Collect** – solid obj. – iron piece, rock salt, stone, wood, key, aluminium
 - Obs. – shape, size – try – hammer
 - Which obj. – particles – strongly held
- Notice – all obj. – solid – **definite shape, volume**
- Reason – particles – tight packed – interparticle attraction – strong
- Strong force – attraction – hold particles – fixed positions
- These particles – to and fro movement – vibration, oscillation – cannot move past each other
- Solids – heated – vibration – more vigorous – stage reached – vibrations – so vigorous – particles leave place – interparticle force – weakened – solid converted – liquid state
- Temp. – this happens – melting pt. – that solid
- Min. temp. – solid melts – become liquid – atm. pressure – **melting point**
- Gen. – liquid state – particles – much farther – compared – solid state
- Ice – exception – particles – much farther – compared – water
- Some solids – weak forces – low melting pt. – others – strong forces – high melting pt.
- Some ex. – solids – melting pt. –
 - Ice – 0 °C
 - Urea – 133 °C
 - Iron – 1538 °C



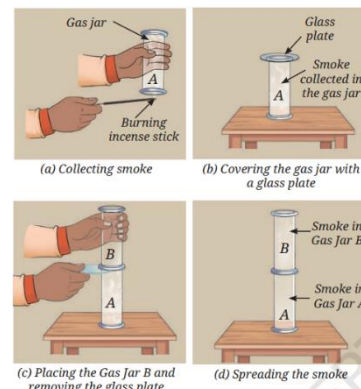
Liquid state

- Activity –
 - Take – 3 clean, dry containers – diff. shapes – label – A, B, C
 - Mark – 200 mL – each container – use marker – OR – thin strip – paper
 - Fill A – water – marked level
 - Transfer water – A to B – no spill – obs. – shape, level – water
 - Transfer water – B to C – no spill – obs. – shape, level – water
- Notice – water – shape – container
- **Liquids** – no fixed shape – take shape – container – reason – particles – free to move
- All 3 containers – water level – 200 mL – no change – obs.
- Liquids – **definite volume** – BUT – container – not clean – some water – stick – wall – next container – water level – slightly less
- Above act. – show – particles – liquids – move freely – BUT – limited space
- **Infer** – liquids – no fixed shape – BUT – fixed volume
- **Compare** – interparticle attraction – liquids, solids
- Take water – shallow vessel – move finger – moving?
- Yes – finger moves – no break, permanent cut – not the same – solids
- Trying this – temporary displacement – water – remove finger – water restored
- Liquids – interparticle attractions – weaker than solids – BUT – strong enough – keep particles close

- Recall – curiosity – grade 6 – chapter – temp. – obs. temp. – boiling water
- Heat liquid – reach stage – start boiling
- Temp. – liquid boils – turns to vapours – atm. pressure – **boiling point**
- Movement – so vigorous – move apart – result – decrease – interparticle attraction
- At a pt. – constituent particles – escape – liquid state – converts to vapour, gaseous state
- Boiling pt. – formation of vapour – very fast – not only surface – BUT also – within liquid – obs. – bubble formation
- Vapour formation – all temp. – even below boiling pt. – slowly – at surface
- Slower process – evaporation – learnt – earlier grades

Gaseous state

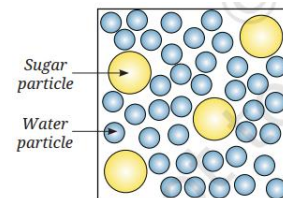
- Activity –
 - Take – 2 transparent gas jars, glass tumblers – mark – A, B
 - Create smoke – burn – incense stick
 - Hold jar A – upside down – over smoke
 - Gas jar – trap smoke – turn over – cover – glass plate
 - Hold jar B – upside down – place over glass plate – jar A
 - Remove glass plate – ensure – both jars – close enough – no gape – smoke escape
 - Obs. – smoke spread – inside jar B
 - Smoke – fill entire jar B – indicate – gases – no fixed volume – occupy entire space – like liquids – acquire shape – vessel
- This shows – gases – free movement – all dir. – interparticle attractions – negligible
- Result – gases – no fixed volume, shape
- This act. – smoke – represent – gaseous state – tiny smoke particles – suspended – air – constantly hit – invisible gas particles – movement – help obs. – motion – gas particle
- This act. – demonstrated – Iodine vapour – obtain – place solid Iodine – closed jar – some time
- Both liquids, gases – flow – do not retain shape – these prop. – distinguish them from solids – **classify** – **fluids**



How Does the Interparticle Spacing Differ in the Three States of Matter?

- Role – interparticle spacing – determine prop. – each state – solid, liquid, gas
- Activity –
 - Take syringe – w/o needle – pull plunger – outwards – fully open
 - Place thumb – open end – prevent inside air – escape
 - Push plunger inward – slowly, steadily – obs.
- Notice – volume of air – inside syringe – decrease
- Compress air – push plunger – particles – forced – come closer
- This shows – gas particles – lots space – natural state – can reduce – apply pressure
- Stop pushing – gas expands – particles spread – plunger moves back
- Repeat act. – use water – obs. – notice – water – incompressible
- Activity –
 - Take – glass vessel – fill half – water – mark water level – A
 - Add 2 teaspoons sugar
 - Mark new level – B
 - Stir water – glass rod – dissolve sugar

- **Predict** – water level – increase, decrease – w.r.t. mark B
- Mark level – C
- Obs. diff. – water levels
- Obs. – initially – sugar added – level increase – BUT – after dissolution – decrease – some extent
- Vol. – solution – less than – vol. – water, sugar – indicate – some space – water particles
- Particles – dissolved substance – occupy this space
- Repeat act. – other soluble solids – common salt, glucose – insoluble solids – sand, stones
- Sand – solid – does not dissolve – water – added to water – settle down – occupy space – total vol. – increase
- Learnt earlier – constituent particles – solids – held together – strong forces
- These particles – no movement – closely packed – BUT – some spacing left – contain nothing
- **A Step Further**
 - Often – term particle – used – diff. contexts – meaning – this term – changes with context
 - Example – air pollution – Suspended Particulate Matter (SPM) – refers – tiny dust particles – suspended – NOT – constituent particles – extremely small
 - Tiny dust particles – made of – large no. – constituent particles – atoms, molecules



How Particles Move in Different States of Matter?

- Activity –
 - Take – glass tumbler – some water – put – few grains – potassium permanganate
 - Obs. – initially – some streaks – pink colour – spreading out
 - Passage of time – entire bulk – uniform pink colour
- Reason – water particles – const. motion – 1st – pull out – potassium permanganate – grain – hit particles – spread through liquid
- Many substances – constituent particles – held together – strong force – water particles – cannot detach them – such substances – sand – insoluble
- Activity –
 - Light – incense stick – 1 corner
 - Wait – few min. – observe – notice fragrance – dist.
- Incense stick – burnt – 1 corner – initially – fragrance – around stick – shortly – feel it – throughout room
- Reason – fragrance particles – spread – fill entire room
- This shows – air particles – const. motion – hit particles – fragrance – help spread – room
- **Interesting fact** –
 - Particulate nature – matter – crucial role – many everyday processes
 - Example – wash oil-stained clothes – use soap – soap particles – surround oil
 - One end – soap particles – attach to oil – other – mix with water – help – lift, wash away – oil
- Based on learning – chapter – matter – made of small particles – held together – attraction forces
- Strength – attractive forces – depend – dist. – in turn – depend – thermal (heat) energy
- Thermal energy – particles – determine – phy. states – matter
- Solid state – thermal energy – min. – remain close – strong interparticle forces – restrict motion – small vibrations
- Melting pt. – thermal energy – overcome attractive forces – allowing state change
- This stage – particles – move away – fixed positions – interparticle dist. – increase – reduce attractive forces – allow particles – move – BUT – limited space

- Gaseous state – particles – enough energy – overcome attractive forces – move freely – all dir.

